

# First overview of governing equations

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## 1 A note on notations

### 1.1 Material derivative

The following equations are all equivalent, which can be shown algebraically.

$$\begin{aligned}\frac{d}{dt} &= \frac{\partial}{\partial t} + u \frac{\partial}{\partial x} + v \frac{\partial}{\partial y} + w \frac{\partial}{\partial z} \\ &= \frac{\partial}{\partial t} + \mathbf{u} \cdot \nabla \\ &= \frac{\partial}{\partial t} + \mathbf{u} \cdot \text{div}()\end{aligned}$$

### 1.2 Tensors

$$\tau_{ij} \frac{\partial u_i}{\partial x_j} = \boldsymbol{\tau} : \nabla \mathbf{u}$$

where

$$\nabla \mathbf{u} = \begin{pmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} & \frac{\partial u}{\partial z} \\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial y} & \frac{\partial v}{\partial z} \\ \frac{\partial w}{\partial x} & \frac{\partial w}{\partial y} & \frac{\partial w}{\partial z} \end{pmatrix}$$

The  $:$  is called the double contraction operator or the matrix inner product. It contracts two tensors into a scalar. Since  $\tau$  is the symmetric stress tensor,

## 2 Starting equations (partial differential equations)

### 2.1 Energy equation

#### 2.1.1 CRB

CRB: Cushman-Roisin and Beckers, 2011

$$\rho C_v \frac{dT}{dt} + p \left( \frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} + \frac{\partial w}{\partial z} \right) = k_T \nabla^2 T \quad (1)$$

I refer to Schubert et al., 2001 for the conversion into the constant-pressure heat capacity variant of this equation.

#### 2.1.2 Jarvis & McKenzie, 1980

jamc80 Jarvis and McKenzie, 1980:

$$\rho C_p \left( \frac{\partial T}{\partial t} - \frac{\alpha T}{\rho C_p} \frac{\partial P}{\partial t} + \mathbf{u} \cdot [\nabla T - (\nabla T)_s] \right) = \nabla \cdot (K \nabla T) + H + \tau_{ij} \frac{\partial u_i}{\partial x_j} \quad (2)$$

### 2.1.3 Gassmüller et al., 2020

Gassmüller: Gassmüller et al., 2020

$$\rho C_p \left( \frac{\partial T}{\partial t} + \mathbf{u} \cdot \nabla T \right) - \nabla \cdot (k \nabla T) = \rho H + \tau : \dot{\varepsilon} + \alpha T (\mathbf{u} \cdot \nabla p) + Q_L \quad (3)$$

### 2.1.4 Schubert et al., 2001

Schubert et al., 2001 presents three thermmal energy equations:

The general form of the energy equation (6.9.1) is:

$$\rho T \frac{Ds}{Dt} = \tau_{ij} \frac{\partial u_i}{\partial x_j} + \frac{\partial}{\partial x_j} \left( k \frac{\partial T}{\partial x_j} \right) + \rho H \quad (4)$$

For an incompressible Newtonian fluid (eq. 6.9.13 + 6.9.14):

$$\rho C_v \frac{DT}{Dt} = \frac{\partial}{\partial x_i} + \frac{\mu}{2} \left( \frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i} \right)^2 + \rho H \quad (5)$$

with  $\frac{\mu}{2} \left( \frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i} \right)^2$  the viscous dissipation function.

In terms of  $C_p$  instead of  $C_v$  (for a compressible fluid) (eq. 6.9.16):

$$\rho C_p \left( \frac{DT}{Dt} - \left( \frac{\partial T}{\partial p} \right)_s \frac{Dp}{Dt} \right) = \frac{\partial}{\partial x_i} + \frac{\mu}{2} \left( \frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i} \right)^2 + \rho H \quad (6)$$

with (eq. 6.9.15, 6.9.17):

$$\begin{aligned} \left( \frac{\partial T}{\partial p} \right)_s &= \frac{\alpha T}{\rho C_p} \\ \left( \frac{DT}{Dt} \right)_s &= \left( \frac{\partial T}{\partial p} \right)_s \frac{Dp}{Dt} = \frac{\alpha T}{\rho C_p} \frac{Dp}{Dt} \end{aligned}$$

## 2.2 Momentum equation

### 2.2.1 CRB

CRB: Cushman-Roisin and Beckers, 2011 per component

$$x : \rho \left( \frac{du}{dt} + f_* w - f v \right) = -\frac{\partial p}{\partial x} + \frac{\partial \tau^{xx}}{\partial x} + \frac{\partial \tau^{xy}}{\partial y} + \frac{\partial \tau^{xz}}{\partial z} \quad (7a)$$

$$y : \rho \left( \frac{dv}{dt} + f u \right) = -\frac{\partial p}{\partial y} + \frac{\tau^{xy}}{\partial x} + \frac{\partial \tau^{yy}}{\partial y} + \frac{\partial \tau^{yz}}{\partial z} \quad (7b)$$

$$z : \rho \left( \frac{dw}{dt} - f_* u \right) = -\frac{\partial p}{\partial z} - \rho g + \frac{\tau^{xz}}{\partial x} + \frac{\tau^{yz}}{\partial y} + \frac{\partial \tau^{zz}}{\partial z} \quad (7c)$$

### 2.2.2 Jarvis & McKenzie, 1980

jamc80, eq. 4 Jarvis and Mckenzie, 1980:

$$\rho \left( \frac{\partial \mathbf{u}}{\partial t} + \mathbf{u} \cdot \nabla \mathbf{u} \right) = -\nabla P + \rho \mathbf{g} + \frac{\partial}{\partial x_j} \tau_{ij} \quad (8)$$

### 2.2.3 Gassmüller et al., 2020

Gassmüller Gassmüller et al., 2020:

$$-\nabla \cdot \tau + \nabla p = \rho \mathbf{g} \quad (9)$$

### 2.2.4 Schubert et al. 2001

$$\rho \frac{du_i}{dt} = -\frac{\partial p}{\partial x_i} + \frac{\partial \tau_{ij}}{\partial x_j} + \rho g_i \quad (10)$$

In 2D, the momentum equation components are:

$$\rho \left( \frac{\partial u_x}{\partial t} + u_x \frac{\partial u_x}{\partial x} + u_y \frac{\partial u_x}{\partial y} \right) = -\frac{\partial p}{\partial x} + \frac{\partial \tau_{xx}}{\partial x} + \frac{\partial \tau_{xy}}{\partial y}, \quad (11)$$

$$\rho \left( \frac{\partial u_y}{\partial t} + u_x \frac{\partial u_y}{\partial x} + u_y \frac{\partial u_y}{\partial y} \right) = -\frac{\partial p}{\partial y} + \frac{\partial \tau_{xy}}{\partial x} + \frac{\partial \tau_{yy}}{\partial y} + \rho g. \quad (12)$$

Note that the assumption is made that gravity acts in the positive y-direction. Section 6.14?

## 2.3 Mass equation (continuity equation) (trivial)

jamc80, eq. 3 Jarvis and Mckenzie, 1980:

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{u}) = 0 \quad (13)$$

CRB Cushman-Roisin and Beckers, 2011:

$$\frac{\partial \rho}{\partial t} + \frac{\partial}{\partial x}(\rho u) + \frac{\partial}{\partial y}(\rho v) + \frac{\partial}{\partial z}(\rho w) = 0 \quad (14)$$

Gassmüller Gassmüller et al., 2020:

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{u}) = 0 \quad (15)$$

schubert et al. (2001) Schubert et al., 2001

$$\frac{\partial \rho}{\partial t} + \frac{\partial}{\partial x_1}(\rho u_i) = \frac{1}{\rho} \frac{D\rho}{Dt} + \frac{\partial u_1}{\partial x_1} = 0 \quad (16)$$

where  $\frac{D}{Dt}$  represents the material derivative (1.1). For incompressible flow,  $\frac{D\rho}{Dt} = 0$ , so eq 18 reduces to:

$$\frac{\partial u_i}{\partial x_i} = \nabla \cdot \mathbf{u} = 0 \quad (17)$$

## 2.4 Starting equations (constitutive equations)

## 2.5 Initial and boundary conditions

# 3 Appendix 1: tables of constants and reference/scaling values used

## 4 Schubert et al., 2001 basic equations

This document/section has a different approach to structure the derivation of the ALA: this is a summary of the derivation of the ALA done by Schubert et al., 2001. **red text means assumptions/approximations were made toward the ALA.**

Table 1: Constants used by Schubert et al., 2001

| Constant | Value | Name                                           |
|----------|-------|------------------------------------------------|
| $k_b$    | 0     | Bulk viscosity                                 |
| $\alpha$ |       | thermal expansivity                            |
| $\chi_T$ |       | isothermal compressibility                     |
| $c_v$    |       | specific heat at constant volume               |
| $c_p$    |       | specific heat at constant pressure             |
| $\chi_a$ |       | adiabatic compressibility                      |
| $k$      |       | thermal conductivity                           |
| $H$      |       | rate of internal heat production per unit mass |
| $K_a$    |       | adiabatic bulk modulus                         |
| $K_T$    |       | isothermal bulk modulus                        |

Table 2: Reference values used by Schubert et al., 2001

| Symbol       | Value                                 | Description                                                      |
|--------------|---------------------------------------|------------------------------------------------------------------|
| $\Delta T$   |                                       | Characteristic temperature difference driving thermal convection |
| $\chi_{T_r}$ |                                       | Representative isothermal compressibility                        |
| $\alpha_r$   |                                       | Representative thermal expansivity                               |
| $u_r$        | $\frac{k_r}{\rho_r c_{p_r} b}$        | Velocity scale                                                   |
| $\mu_r$      | $10^{21}$ Pa s                        | Representative viscosity                                         |
| $\chi_a$     |                                       | Adiabatic compressibility                                        |
| $b$          |                                       | Depth of convecting region                                       |
| $p_r$        | $\frac{\mu_r u_r}{b}$                 | Scale for dynamic pressure variations                            |
| $k_r$        | $4 \text{ W m}^{-1} \text{ K}^{-1}$   | Reference value thermal conductivity                             |
| $c_{p_r}$    | $1 \text{ kJ kg}^{-1} \text{ K}^{-1}$ | Reference specific heat at constant pressure                     |

#### 4.1 Mass equation/continuity equation/conservation of mass

$$\frac{\partial \rho}{\partial t} + \frac{\partial}{\partial x_1}(\rho u_i) = \frac{1}{\rho} \frac{D\rho}{Dt} + \frac{\partial u_1}{\partial x_1} = 0 \quad (18)$$

where  $\frac{D}{Dt}$  represents the material derivative (1.1). For incompressible flow,  $\frac{D\rho}{Dt} = 0$ , so eq 18 reduces to:

$$\frac{\partial u_i}{\partial x_i} = \nabla \cdot \mathbf{u} = 0 \quad (19)$$

For the special case of an incompressible fluid,  $\frac{D\rho}{Dt} = 0$ , so equation 19 reduces to

$$\frac{\partial u_i}{\partial x_i} = 0 \quad (20)$$

#### 4.2 Momentum equations/conservation of momentum

Starting governing equation (6.4.1):

$$\rho \frac{Du_i}{Dt} = -\frac{\partial p}{\partial x_i} + \frac{\partial \tau_{ij}}{\partial x_j} + \rho g_i \quad (21)$$

Table 3: Dimensionless parameters and their values for the Earth's mantle

| Parameter      | Definition                                      | Value                | Description                                             |
|----------------|-------------------------------------------------|----------------------|---------------------------------------------------------|
| Pr             | $\frac{\mu_r c_{p_r}}{k_r}$                     | $2.5 \times 10^{23}$ | Prandtl number                                          |
| M <sup>2</sup> | $\frac{k_r^2 \chi_{T_r}}{\rho_r c_{p_r}^2 b^2}$ | $10^{-33}$           | Mach number (squared)                                   |
| $\epsilon$     | $\alpha_r \Delta T_r$                           |                      | Fractional density change due to temperature variations |

we assume gravity  $g$  to be the only body force. The coriolis force is negligibly small in the Earth's mantle, and the centrifugal force due to the Earth's rotation is included in the gravity term.

In 2D, the momentum equation components are:

$$\rho \left( \frac{\partial u_x}{\partial t} + u_x \frac{\partial u_x}{\partial x} + u_y \frac{\partial u_x}{\partial y} \right) = -\frac{\partial p}{\partial x} + \frac{\partial \tau_{xx}}{\partial x} + \frac{\partial \tau_{xy}}{\partial y}, \quad (22)$$

$$\rho \left( \frac{\partial u_y}{\partial t} + u_x \frac{\partial u_y}{\partial x} + u_y \frac{\partial u_y}{\partial y} \right) = -\frac{\partial p}{\partial y} + \frac{\partial \tau_{xy}}{\partial x} + \frac{\partial \tau_{yy}}{\partial y} + \rho g. \quad (23)$$

Note that the assumption is made that gravity acts in the positive y-direction. Section 6.14?ext

### 4.3 Navier-Stokes equation

The local strain rate tensor  $e_{ij}$  is given by:

$$e_{ij} = \frac{1}{2} \left( \frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i} \right) \quad (24)$$

For a Newtonian, isotropic fluid, the relation between the deviatoric(!) stress tensor and the strain rate tensor is

$$\tau_{ij} = 2\mu e_{ij} + \lambda e_{kk} \delta_{ij} \quad (25)$$

Defining the bulk viscosity  $k_B$ :

$$\frac{\tau_{ii}}{3} = e_{ii} \left( \lambda + \frac{2}{3} \mu \right) = k_B e_{ii} \quad (26)$$

The combination of equations 25, 26 and 24 gives

$$\tau_{ij} = \mu \left( \frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i} \right) + \left( k_B - \frac{2}{3} \mu \right) \frac{\partial u_k}{\partial x_k} \delta_{ij} \quad (27)$$

and substitution of eq. 27 into eq. 21 gives

$$\rho \frac{Du_i}{Dt} = -\frac{\partial p}{\partial x_i} + \frac{\partial}{\partial x_j} \left[ \mu \left( \frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i} \right) + \left( k_B - \frac{2}{3} \mu \right) \frac{\partial u_k}{\partial x_k} \delta_{ij} \right] + \rho g_i \quad (28)$$

For many fluids,  $k_B$  is taken to be zero (**Stokes assumption**), so eq. 27 becomes

$$\tau_{ij} = \mu \left( \frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i} - \frac{2}{3} \frac{\partial u_k}{\partial x_k} \delta_{ij} \right) \quad (29)$$

and eq. 28 becomes the Navier-Stokes equation:

$$\rho \frac{Du_i}{Dt} = -\frac{\partial p}{\partial x_i} + \frac{\partial}{\partial x_j} \left[ \mu \left( \frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i} - \frac{2}{3} \frac{\partial u_k}{\partial x_k} \delta_{ij} \right) \right] + \rho g_i \quad (30)$$

for incompressible flow, equation 19 tells us  $\frac{\partial u_k}{\partial x_k} = 0$ , and together with a constant dynamic viscosity  $\mu$  so eq. 30 reduces to:

$$\rho \frac{Du_i}{Dt} = -\frac{\partial p}{\partial x_i} + \mu \frac{\partial^2 u_i}{\partial x_j^2} + \rho g_i \quad (31)$$

This equation represents a balance between inertial forces  $\left(\frac{Du_i}{Dt}\right)$ , pressure forces  $\left(\frac{\partial p}{\partial x_i}\right)$ , viscous forces  $\left(\mu \frac{\partial^2 u_i}{\partial x_j^2}\right)$  and the body force due to gravity  $(\rho g_i)$

#### 4.4 Thermal energy equations/conservation of energy

Schubert et al., 2001 presents various different formulations of the energy equation.

The general form of the energy equation (6.9.1) is:

$$\rho T \frac{Ds}{Dt} = \tau_{ij} \frac{\partial u_i}{\partial x_j} + \frac{\partial}{\partial x_j} \left( k \frac{\partial T}{\partial x_j} \right) + \rho H \quad (32)$$

The terms on the right side of 32 are, respectively, volumetric heat production rates due to viscous dissipation (internal friction), thermal conduction, and internal heat generation (radiogenic?). **Where it is assumed that Fourier's law of heat conduction for an isotropic medium is applicable:**

$$q_i = -k \frac{\partial T}{\partial x_i} \quad (33)$$

Where  $q_i$  is the heat flux vector. Additionally, the viscous dissipation term can be written as the dissipation function:

$$\Phi = \tau_{ij} \frac{\partial u_i}{\partial x_j} \quad (34)$$

Inserting the definition of our deviatoric stress tensor with  $k_B = 0$  (eq. ??), this becomes

$$\Phi = 2\mu \left\{ \frac{1}{2} \left( \frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i} \right) - \frac{1}{3} \left( \frac{\partial u_k}{\partial x_k} \right) \delta_{ij} \right\}^2 \quad (35)$$

**For an incompressible and Newtonian fluid ( $\frac{\partial u_i}{\partial x_i} = 0$ ), the dissipation function reduces to**

$$\Phi = \frac{\mu}{2} \left( \frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i} \right)^2 \quad (36)$$

With various thermodynamic relationships:

$$\alpha = \frac{1}{v} \left( \frac{\partial v}{\partial T} \right)_p \quad (37)$$

$$c_p = T \left( \frac{\partial s}{\partial T} \right)_p \quad (38)$$

$$\left( \frac{\partial s}{\partial p} \right)_T = - \left( \frac{\partial v}{\partial T} \right)_p \quad (39)$$

It can be shown that

$$\frac{Ds}{Dt} = \frac{c_p}{T} \frac{DT}{Dt} - \frac{\alpha}{\rho} \frac{Dp}{Dt} \quad (40)$$

Plugging 40 into 32 gives

$$\begin{aligned} \rho c_p \frac{DT}{Dt} - \alpha T \frac{DP}{Dt} &= \frac{\partial}{\partial x_i} \left( k \frac{\partial T}{\partial x_i} \right) + \Phi + \rho H \\ \rho c_p \left( \frac{DT}{Dt} - \frac{\alpha T}{\rho c_p} \frac{DP}{Dt} \right) &= \frac{\partial}{\partial x_i} \left( k \frac{\partial T}{\partial x_i} \right) + \Phi + \rho H \end{aligned} \quad (41)$$

## 5 Approximate equations

In this section, Schubert derives the approximate forms of the equations of conservation of energy and momentum appropriate to mantle convection.

### 5.1 Equation of state (density $\rho$ )

From now on, density  $\rho$  is considered a function of temperature  $T$  and pressure  $p$  only, neglecting the density changes associated with mineral phase transitions and changes in chemical composition.

Most of the density change in the mantle is due to hydrostatic compression. The mantle density increases by approximately 65% between the surface and the core-mantle boundary. This is Schubert's justification for assuming a linearized equation of state for mantle density:

$$\begin{aligned}\rho &= \bar{\rho}(\bar{T}, \bar{p}) + \rho' \\ \rho &= \bar{\rho}(\bar{T}, \bar{p}) + \left(\frac{\partial \rho}{\partial p}\right)_T p' + \left(\frac{\partial \rho}{\partial T}\right)_p T' \\ \rho &= \bar{\rho}(\bar{T}, \bar{p}) + \bar{\rho}\chi_T p' - \bar{\alpha}\rho T'\end{aligned}\tag{42}$$

with

$$\begin{aligned}T &= \bar{T} + T' \\ p &= \bar{p} + p'\end{aligned}\tag{43}$$

Where the overbar ( $\bar{\rho}$ ) refers to a reference state and the primes ( $\rho'$ ) departures from it. The reference state is assumed motionless and steady:

- motionless:  $\bar{u}_i = 0$
- steady:  $\frac{\partial \bar{u}_i}{\partial t} = 0$
- motionless and steady:  $\frac{D\bar{u}_i}{Dt} = 0$
- $\tau_{ij} = \mu \left( \frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i} - \frac{2}{3} \frac{\partial u_k}{\partial x_k} \delta_{ij} \right) = 0$  (29)

This reduces the momentum equation (30, 21) to its static form:

$$0 = -\frac{\partial \bar{p}}{\partial x_i} + \bar{\rho}g_i\tag{44}$$

Integrating equation 44 results in the hydrostatic equation. Further needed for the calculation of the reference state is the equation for the adiabatic temperature gradient (since the reference state temperature is assumed adiabatic)

$$\left(\frac{\partial T}{\partial x_i}\right)_s = \frac{\alpha T g_i}{c_p}\tag{45}$$

And a specification of the reference state equation of state  $\bar{\rho} = \bar{\rho}(\bar{T}, \bar{p})$ .

### 5.2 Nondimensionalization

Now, the linearized equation of state (42) will be nondimensionalized with scaling factors consisting of representative/characteristic values presented in table 5. With these, dimensionless variables (denoted with an asterisk  $*$ ) are introduced by dividing the dimensional variables with their characteristic counterpart. Rewriting the linearized equation of state (42) in terms of these nondimensional variables gives:

$$\frac{\rho^*}{\rho^*} = 1\tag{46}$$

Table 4: Constants used by Schubert et al., 2001

| Constant | Value | Name                                           |
|----------|-------|------------------------------------------------|
| $k_b$    | 0     | Bulk viscosity                                 |
| $\alpha$ |       | thermal expansivity                            |
| $\chi_T$ |       | isothermal compressibility                     |
| $c_v$    |       | specific heat at constant volume               |
| $c_p$    |       | specific heat at constant pressure             |
| $\chi_a$ |       | adiabatic compressibility                      |
| $k$      |       | thermal conductivity                           |
| $H$      |       | rate of internal heat production per unit mass |
| $K_a$    |       | adiabatic bulk modulus                         |
| $K_T$    |       | isothermal bulk modulus                        |

Table 5: Reference values used by Schubert et al., 2001

| Symbol       | Value                                          | Description                                                      |
|--------------|------------------------------------------------|------------------------------------------------------------------|
| $\Delta T_r$ |                                                | Characteristic temperature difference driving thermal convection |
| $\chi_{T_r}$ |                                                | Representative isothermal compressibility                        |
| $\alpha_r$   |                                                | Representative thermal expansivity                               |
| $u_r$        | $\frac{k_r}{\rho_r c_{p_r} b}$                 | Velocity scale (follows from assuming $Pe = 1$ )                 |
| $\mu_r$      | $10^{21}$ Pa s                                 | Representative viscosity                                         |
| $b$          |                                                | Depth of convecting region                                       |
| $p_r$        | $\frac{\mu_r u_r}{b}$                          | Scale for dynamic pressure variations                            |
| $k_r$        | $4 \text{ W m}^{-1} \text{ K}^{-1}$            | Reference value thermal conductivity                             |
| $c_{p_r}$    | $1 \text{ kJ kg}^{-1} \text{ K}^{-1}$          | Reference specific heat at constant pressure                     |
| $t_r$        | $\frac{b}{u_r} = \frac{\rho_r c_{p_r} b^2}{k}$ | reference timescale                                              |

## 6 Appendix 1: tables of constants and reference/scaling values used

### References

- Cushman-Roisin, B., & Beckers, J.-M. (2011). *Introduction to geophysical fluid dynamics: Physical and numerical aspects*. Academic Press.
- Gassmüller, R., Dannberg, J., Bangerth, W., Heister, T., & Myhill, R. (2020). On formulations of compressible mantle convection. *Geophysical Journal International*, 221(2), 1264–1280. <https://doi.org/10.1093/gji/ggaa078>
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- Schubert, G., Turcotte, D. L., & Olson, P. (2001). *Mantle convection in the earth and planets*. Cambridge University Press.

Table 6: Dimensionless parameters and their values for the Earth’s mantle

| Parameter      | Definition                                      | Value                | Description                                             |
|----------------|-------------------------------------------------|----------------------|---------------------------------------------------------|
| Pr             | $\frac{\mu_r c_{p_r}}{k_r}$                     | $2.5 \times 10^{23}$ | Prandtl number                                          |
| M <sup>2</sup> | $\frac{k_r^2 \chi_{T_r}}{\rho_r c_{p_r}^2 b^2}$ | $10^{-33}$           | Mach number (squared)                                   |
| $\epsilon$     | $\alpha_r \Delta T_r$                           |                      | Fractional density change due to temperature variations |